Short-pulse sequences as local probes for superconducting-qubit frequency calibration

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Qubit calibration normally extracts frequency information from dedicated spectroscopic scans or free evolution between control pulses. We investigate whether a short-pulse sequence can instead contribute directly to calibration through the detuning response accumulated during the pulses themselves. As a case study, two orthogonal, zero-delay Ramsey sequences form a local frequency discriminator whose first- and third-order response kernels map population to detuning. The cubic correction requires the full three-time kernel rather than its time-diagonal restriction. The short-pulse probe is only locally invertible, so its use in a feedback loop additionally requires the drive reference to track the predicted qubit frequency and keep each measurement inside the calibrated response window. In numerical closed-loop tests for a 20-MHz target shift, the resulting protocol reaches a 5.2-kHz independently verified residual in six iterations and 984 counted solver calls, compared with an 18.0-kHz residual and 4000 calls for a double-sweep Ramsey baseline. These results establish a concrete regime in which short-pulse sequences can participate in local calibration, while also identifying response folds and reference tracking as the constraints that delimit that role.

I. INTRODUCTION

Superconducting qubits require frequent calibration because transition frequencies drift with flux bias, temperature, control conditions, and the electromagnetic environment. Frequency error appears directly as a detuning in single-qubit control and propagates into gate, readout, and higher-level calibration tasks [? ?]. Modern calibration systems therefore organize repeated measurements and parameter updates into automated feedback loops [? ?]. Reducing the cost of the elementary frequency measurement is valuable because the same primitive is executed across devices and repeated as the processor drifts [? ?]. This motivates a broader question: can the short control-pulse sequences already used around a working point provide calibration information themselves?

Ramsey spectroscopy is a robust frequency estimator. A pair of $\pi/2$ pulses separated by a variable free-evolution interval converts detuning into population oscillations, and the oscillation frequency is extracted from a scan. The scan provides a useful capture range and high precision, but its cost scales with the number of delay points and, when two artificial detunings are used to determine the sign, with two complete scans. Fixed-delay Ramsey measurements reduce this burden once the frequency is already known locally, and have been incorporated into closed-loop stabilization protocols [?]. Their local phase response, however, becomes ambiguous after phase wrapping.

Here we test this possibility in a frequency-calibration setting. We remove the free-evolution interval and infer detuning from the response accumulated during two finite control pulses. Orthogonal analysis branches form an odd differential signal whose slope is determined by a response kernel. Such control-time response functions connect temporal sensing to driven-qubit dynamics [? ?]. They suggest that a short pulse can be more than an operation surrounding a calibration experiment: its transient response can itself be the probe. The question is then not only whether this response contains frequency information, but whether that local information can remain valid as a calibration loop moves the device.

Two constraints are central. First, the short-pulse response is nonlinear and only locally invertible; its leading correction depends on the full three-time response kernel. Second, if the feedback loop changes the flux bias while the drive remains fixed, the qubit can leave the discriminator's monotonic window, corrupting both the frequency estimate and the subsequent controller update.

Our central result is that short-pulse sequences can participate in closed-loop frequency calibration when these two constraints are controlled. A full three-time response kernel supplies the local cubic inverse. A secant estimate of the frequency–flux slope then predicts the transition frequency at the next bias point. Driving at this predicted frequency keeps the pulse response inside its calibrated window, while the feedback residual remains defined relative to the fixed target. The response kernel and drive tracking therefore play supporting but distinct roles: one extracts information from the short sequence, and the other preserves the validity of that information along the calibration trajectory.

We first derive the short-pulse discriminator and its local inverse. We then separate drive tracking, which preserves measurement validity, from the flux update, which closes the control loop. Numerical tests compare the local probe with Ramsey strategies using independently verified frequency errors and instrumented solver counts. Besides demonstrating a useful operating regime, the comparison exposes a negative result: simply placing a short-pulse method before Ramsey in a coarse-to-fine workflow can cost more than Ramsey alone.

II. SYSTEM AND CALIBRATION TASK

A. Flux-tunable transmon and calibration objective

We consider a flux-tunable transmon qubit [?] controlled through a calibrated flux-bias coordinate Φ. Experimentally, Φ is controlled through a room-temperatue bias voltage V_b using a separately characterized voltage-to-flux conversion, as detailed in the Supplemental Material.

In the transmon regime, the flux dispersion of the qubit angular transition frequency is approximately given by [?]

$$\begin{aligned} \hbar\omega_{01}(\Phi) &\simeq \sqrt{8E_C E_J(\Phi)} - E_C, \\ E_J(\Phi) &= E_{J,\Sigma}\sqrt{\cos^2(\pi\Phi/\Phi_0) + d^2\sin^2(\pi\Phi/\Phi_0)}, \end{aligned} \tag{1}$$

Equation (1) describes the flux dependence of the transition frequency and thus provides the physical basis for frequency calibration through flux-bias tuning. The calibration objective is to find a flux bias $\Phi_\star$ such that

$$|\omega_{01}(\Phi_\star) - \omega_\mathrm{tar}| \le \epsilon_\omega, \tag{2}$$

where ω_tar is the target angular transition frequency and ϵ_ω is the prescribed tolerance.

In practice, a previously characterized frequency–flux map may no longer be accurate at the target tolerance because of slow shifts in the effective flux offset or device parameters[? ?]. We therefore formulate the calibration without assuming an up-to-date quantitative map $\omega_{01}(\Phi)$: the controller operates in the calibrated flux coordinate and obtains the local frequency information required for bias updates directly from measurements.

Figure 1(a) summarizes this calibration setting. The full dispersion illustrates the physical tunability of the qubit, while the inset shows a local trajectory from the seed bias toward the target frequency. The displayed dispersion is withheld from the controller, which uses only the frequency information obtained at the visited bias points.

B. Ramsey and bipartite short-pulse probes

At each calibration iteration, the qubit is interrogated at a fixed flux bias by a control sequence followed by population readout. We refer to this pulse-and-readout operation as a measurement probe. We consider two types of probes: Ramsey probes and bipartite short-pulse probes. The driven dynamics of both probes are described by an effective two-level Hamiltonian. In a frame rotating at the drive frequency ω_d and under the rotating-wave approximation [? ?],

$$\begin{aligned} \frac{H(t,\Phi)}{\hbar} &= \frac{\Delta(\Phi)}{2}\sigma_z + \frac{\Omega(t)}{2}\left[\cos\varphi(t)\sigma_x + \sin\varphi(t)\sigma_y\right], \\ \Delta(\Phi) &= \omega_{01}(\Phi) - \omega_d. \end{aligned} \tag{3}$$

Here $\Delta(\Phi)$ is the angular-frequency detuning, while $\Omega(t)$ and $\varphi(t)$ are the drive envelope and phase. The detuning term connects the unknown transition frequency to the driven dynamics. Ramsey and short-pulse probes use different control sequences to infer this detuning from population readout; once $\widehat{\Delta}$ is obtained, the transition frequency follows as $\widehat{\omega}_{01} = \omega_d + \widehat{\Delta}$.

Figure 1(b) compares the timing structures of the two probe types. We first consider the Ramsey probe, which uses two nominal $\pi/2$ pulses separated by a free-evolution interval τ [?].

For a Ramsey sequence with free-evolution time τ and analysis phase φ, the excited-state population is

$$p_R(\Delta;\tau,\varphi) = \frac{1}{2}\left[1 + C\cos(\Delta\tau + \varphi)\right], \tag{4}$$

where C is the Ramsey contrast. We use this response in two measurement modes. Scanned Ramsey samples multiple values of τ to estimate the oscillation frequency over a broad range. Once the detuning has been confined to a known local interval, τ can instead be fixed at τ_0, and two orthogonal analysis phases recover detuning from the measured quadratures, subject to phase wrapping [?]. These modes provide a wide-range frequency estimate and a local frequency-estimation baseline, respectively. The scan processing, phase reconstruction, and choice of τ_0 are specified in Appendix A.

We next turn to the bipartite short-pulse probe. Unlike the Ramsey probe, it contains no separate free-evolution interval and instead applies a common preparation rotation of angle α immediately followed by one of two analysis rotations whose phases differ by π [?]:

$$R_y(\alpha) \longrightarrow R_{+x}(\alpha) \quad \text{or} \quad R_y(\alpha) \longrightarrow R_{-x}(\alpha). \tag{5}$$

Here $\alpha = \int \Omega(t)\,\mathrm{d}t$ is determined from the pulse envelope and duration. Varying α changes the differential detuning response and therefore the response coefficients and validated operating range introduced in Sec. III.

We refer to these two sequences as the $+X$ and $-X$ branches and denote their final excited-state populations by p_{+X} and p_{-X}, respectively. We define the corresponding differential observable as

$$p_\mathrm{d} = \frac{p_{+X} - p_{-X}}{2}. \tag{6}$$

Figure 1(c) shows how the two branch populations form the differential response. Detuning nevertheless acts during the finite control pulses and changes the measured branch populations. Their difference provides a first-order-sensitive local frequency observable when

$$\left.\frac{\partial p_\mathrm{d}}{\partial \Delta}\right|_{\Delta=0} \neq 0. \tag{7}$$

This condition is necessary for local inversion but does not guarantee an unambiguous response over an unrestricted detuning range. Section III quantifies the local response and establishes its validated operating interval.

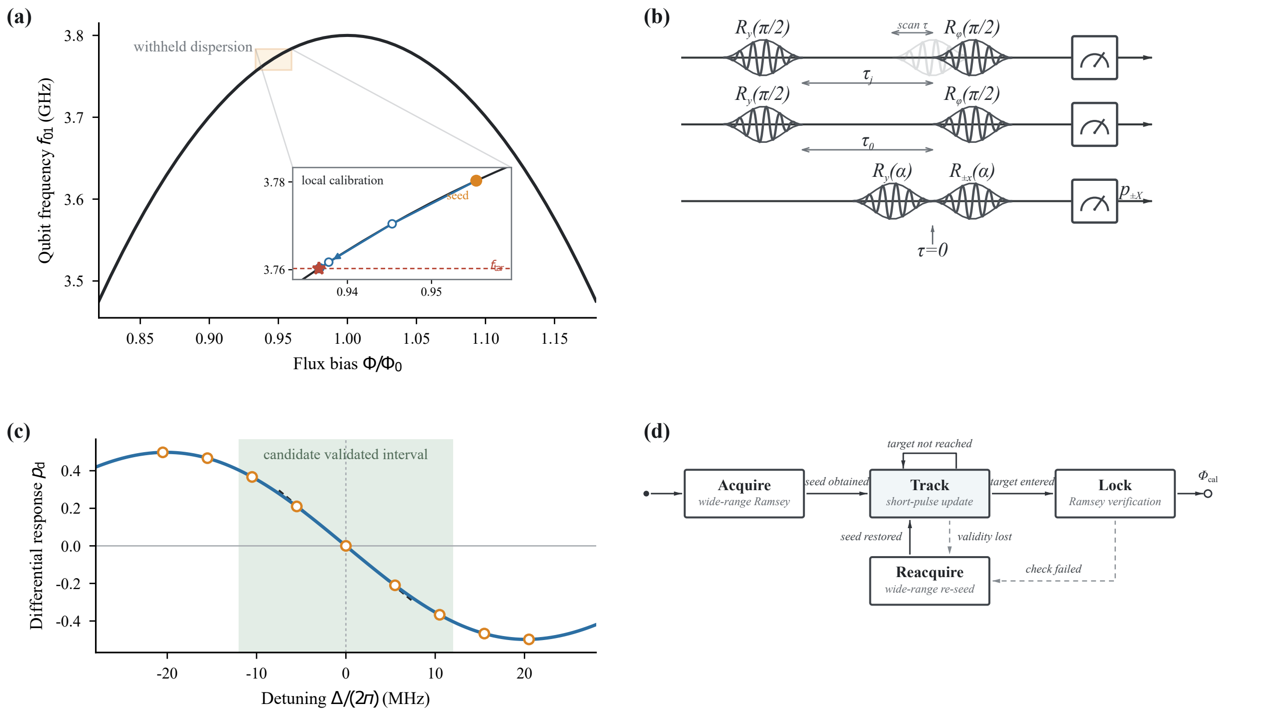

FIG. 1. Schematic overview of the physical system, measurement probes, and calibration protocol. (a) Qubit frequency dispersion, with an inset showing the seed bias, target frequency, and successive points of the local calibration trajectory. The complete dispersion is withheld from the calibration controller. (b) Timing diagrams for scanned Ramsey, fixed-delay Ramsey, and the bipartite short-pulse probe. (c) Construction of the differential observable p_{d} from the phase-cycled $+X$ and $-X$ branches, together with its locally monotonic response about resonance. (d) The ACQUIRE–TRACK–LOCK–REACQUIRE protocol, distinguishing flux-bias updates from drive-reference updates.

C. Calibration workflow and independent validation

The local-identifiability condition above concerns a probe at a fixed bias. During calibration, however, bias updates can move the qubit frequency beyond the invertible range of the short-pulse response. We therefore use the four-state ACQUIRE–TRACK–LOCK–REACQUIRE workflow shown in Fig. 1(d), combining wide-range frequency acquisition and recovery with local tracking and independent target confirmation. This organization follows the general setting of automated qubit calibration and closed-loop frequency feedback [? ? ?].

Measurement, estimation, and tuning remain distinct within the workflow. A probe returns population data, an estimator converts them into a frequency estimate and its uncertainty, and the tuning controller updates the physical flux bias. A separate drive-reference update changes only the reference of the next measurement. These two updates are governed by different frequency differences:

$$\Delta_n = \omega_{01}(\Phi_n) - \omega_{d,n}, \qquad r_n = \omega_{01}(\Phi_n) - \omega_{\mathrm{tar}}. \tag{8}$$

The measurement detuning Δ_n determines whether the local probe remains valid, whereas the calibration residual r_n determines whether the target has been reached. The flux-bias update reduces r_n by changing the physical working point, while the drive-reference update controls Δ_n without moving that point.

In LOCK, the drive is placed at ω_{tar}, and an independent frequency measurement tests

$$\left|\widehat{\omega}_{01}^{(\mathrm{ver})} - \omega_{\mathrm{tar}}\right| + z_{1-\beta}\sigma_{\mathrm{ver}} \leq \epsilon_\omega. \tag{9}$$

Calibration success is declared only after N_{lock} consecutive verification measurements satisfy Eq. (9). A failed verification returns the protocol to TRACK if the local-entry conditions remain satisfied and to REACQUIRE otherwise. Thus, LOCK provides independent target confirmation and short-term holding rather than continued flux refinement or long-term stabilization.

REACQUIRE repeats the wide-range frequency measurement and returns a new bias–frequency seed together with its uncertainty. A successful local-entry test then starts a new TRACK episode. The numerical thresholds for the Ramsey capture range, confidence margins, sensitivity checks, repeated-failure rule, and lock confirmation are specified in Appendix A.

To avoid self-validation, the verification measurement is independent of the short-pulse estimator used inside the feedback loop. We define $T_{\mathrm{hit}}(\epsilon_\omega)$ as the elapsed experimental time at which Eq. (9) is first confirmed with the prescribed confidence. The verification schedule and the cold-start, amortized, and local-tracking cost conventions are given in Appendix A.

Section III now constructs the estimator and controller that implement the local TRACK state. The wide-range acquisition, independent lock verification, and state-transition roles remain those defined here.

III. SHORT-PULSE FREQUENCY INFERENCE AND CALIBRATION

We now specify the local TRACK branch introduced in Sec. II.C. A usable tracking iteration must convert the differential population into a frequency estimate, move the physical working point toward ω_{tar}, and keep the next measurement within the validated response interval as the qubit frequency changes. We therefore derive the short-pulse differential response, construct its nonlinear local inverse together with an explicit validity boundary, use the resulting frequency estimate in a bounded flux-bias update, and update the drive reference to preserve the local measurement condition. Together, these elements specify how, and within what limits, short-pulse frequency information can be incorporated into closed-loop flux-bias calibration.

A. Short-pulse differential response

To characterize how detuning is encoded in Eq. (6), we regard the differential population as a functional of a time-dependent detuning $\Delta(t)$. Following the general framework of linear and nonlinear response theory [**?** **?**], we expand the differential population around the resonant control evolution as

$$p_{\mathrm{d}}[\Delta] = p_{\mathrm{d}}^{(0)} + \sum_{m=1}^{\infty} \frac{1}{m!} \int_0^T \mathrm{d}t_1 \cdots \int_0^T \mathrm{d}t_m \times k_m(t_1, \ldots, t_m) \prod_{j=1}^{m} \Delta(t_j). \tag{10}$$

Here $k_m(t_1, \ldots, t_m)$ is the symmetrized mth-order detuning-response kernel of the complete bipartite short-pulse probe. It quantifies the joint contribution of detuning at the times $t_1, \ldots, t_m$ to the final differential population, and T is the total sequence duration. The response-function derivation, including time ordering and nested commutators, is given in the Supplemental Material.

For scalar local frequency estimation, we restrict attention to the quasi-static regime in which detuning variations during a single short sequence are negligible, so that $\Delta(t) \simeq \Delta$. This approximation does not preclude drift between repetitions or feedback iterations.

For the ideal $+X/-X$ probe defined in Eq. (5), detuning reversal interchanges the branch responses, $p_{+X}(-\Delta) = p_{-X}(\Delta)$. The differential combination therefore satisfies $p_{\mathrm{d}}(-\Delta) = -p_{\mathrm{d}}(\Delta)$. In particular, $p_{\mathrm{d}}^{(0)} = 0$, and all even-order terms in Eq. (10) vanish. The local response can thus be written as

$$p_{\mathrm{d}}(\Delta) = G_1 \Delta + \frac{G_3}{3!} \Delta^3 + \mathcal{O}(\Delta^5), \tag{11}$$

where G_1 and G_3 are the first and third derivatives of p_{d} with respect to Δ at resonance. The coefficient G_1 determines the local sensitivity, while G_3 gives the leading nonlinear correction.

For the experimental implementation, we separate the response measurements into calibration and validation data. From the response-calibration data, we obtain G_1 and G_3 through an offline, measurement-based kernel calibration. Calibrated Virtual-Z phase insertions are applied at selected times within the two control branches, and finite differences of the resulting branch populations sample the detuning-response derivatives. In the quasi-static limit, the integrated coefficients are

$$\begin{aligned} G_1 &= \int_0^T k_1(t)\ \mathrm{d}t, \\ G_3 &= \iiint_{[0,T]^3} k_3(t_1, t_2, t_3)\ \mathrm{d}t_1\, \mathrm{d}t_2\, \mathrm{d}t_3. \end{aligned} \tag{12}$$

The Virtual-Z operations follow Ref. [**?**]. The finite-difference construction and three-time sampling scheme used to obtain the response kernels are detailed in the Supplemental Material.

Experimental imperfections may nevertheless produce a nonzero measured differential signal at resonance. We therefore define

$$\widetilde{p}_{\mathrm{d}} = p_{\mathrm{d}}^{\mathrm{meas}} - b_0, \tag{13}$$

where b_0 is the zero-detuning intercept estimated from the response-calibration data and held fixed thereafter. This operation corrects a differential-population offset; it does not shift the frequency reference.

For comparison, an odd-polynomial fit to the corrected quasi-static response provides a separate check of the integrated coefficients but is not used to generate the coefficients supplied to the main feedback protocol. Once G_1, G_3, and b_0 have been fixed, separately acquired response-validation data are used to establish Δ_{val} and to assess residual symmetry breaking through $p_{\mathrm{even}}(\Delta) = [\widetilde{p}_{\mathrm{d}}(\Delta) + \widetilde{p}_{\mathrm{d}}(-\Delta)]/2$. The even component is treated as a diagnostic of model mismatch rather than absorbed into the ideal odd response model. This response validation is distinct from the independent frequency verification used to confirm calibration success in Sec. II.C.

The acquisition and partitioning of the calibration and validation data, together with the sampling schedule, coefficient-reuse range, recalibration conditions, and associated cost accounting, are specified in Appendix A.

B. Local frequency estimator

The first-order frequency estimate follows from the local slope:

$$\widehat{\Delta}^{(1)} = \frac{\widetilde{p}_{\rm d}}{G_1}, \qquad \widehat{\omega}_{01}^{(1)} = \omega_d + \widehat{\Delta}^{(1)}. \tag{14}$$

Its truncation bias increases as the probe moves away from resonance. This estimate provides the initial value for the cubic inversion and the first-order reference used to quantify nonlinear-estimation bias in Sec. IV.

The main local estimator retains the third-order response and solves

$$G_1\widehat{\Delta}^{(3)} + \frac{G_3}{6}\left(\widehat{\Delta}^{(3)}\right)^3 = \widetilde{p}_{\rm d}, \tag{15}$$

with $\widehat{\omega}_{01}^{(3)} = \omega_d + \widehat{\Delta}^{(3)}$. Newton iteration is initialized with $\widehat{\Delta}^{(1)}$, and the candidate solution is selected from the branch connected continuously to $\Delta = 0$.

When $G_1G_3 < 0$, the cubic response has stationary points at

$$|\Delta_{\rm fold}| = \sqrt{-\frac{2G_1}{G_3}}. \tag{16}$$

Beyond these points, the cubic response is no longer one-to-one, and the same differential population can correspond to multiple detunings. The response fold is therefore a mathematical boundary of the truncated cubic model, not an experimentally established operating range.

We define the validated local operating interval by

$$|\Delta| \le \Delta_{\rm val}, \qquad \Delta_{\rm val} < |\Delta_{\rm fold}| \quad (G_1G_3 < 0). \tag{17}$$

The value of $\Delta_{\rm val}$ is established from the held-out response-validation data using predefined estimator-error and confidence criteria. It remains necessary when the cubic model has no real fold because model truncation, measurement noise, and experimental imperfections can restrict the usable response before a mathematical turnover occurs. Failure of numerical convergence, branch continuity, or the validated-range check rejects the estimate and triggers REACQUIRE rather than extrapolation of the local inverse.

C. Track initialization and flux-bias update

Section II.C defined the state transitions of the complete calibration workflow. Here we construct the local controller that operates after an ACQUIRE or REACQUIRE measurement has supplied the seed

$$(\Phi_{\rm seed}, \widehat{\omega}_{01,\rm seed}, \sigma_{\omega,\rm seed}). \tag{18}$$

The acquired frequency estimate is reused rather than repeated with the short-pulse probe at the same bias. It initializes the local drive reference as $\omega_{d,\rm seed} = \widehat{\omega}_{01,\rm seed}$ and gives the seed residual

$$\widehat{r}_{\rm seed} = \widehat{\omega}_{01,\rm seed} - \omega_{\rm tar}. \tag{19}$$

If Eq. (**??**) is already satisfied, the workflow proceeds directly to LOCK. Otherwise, the seed is used to initialize the local flux-bias update.

Because one bias–frequency pair does not determine a secant sensitivity, the controller proposes a bounded one-sided step,

$$\Phi_1^{\rm prop} = \Phi_{\rm seed} - \chi_\Phi \,\mathrm{sgn}(\widehat{r}_{\rm seed})\delta\Phi_{\rm blind}, \tag{20}$$

where $\chi_\Phi = \mathrm{sgn}(\partial\omega_{01}/\partial\Phi) \in \{-1, +1\}$ specifies the known local monotonic direction. The applied bias Φ_1 is restricted to the allowed interval $[\Phi_{\min}, \Phi_{\max}]$.

Let $\Delta_{\rm blind}^{\max}$ denote an experimentally established upper bound on the frequency displacement produced by the applied blind step. The proposed initialization is admitted to TRACK only if

$$z_{1-\beta}\sigma_{\Delta,\rm seed} + \Delta_{\rm blind}^{\max} \le \Delta_{\rm val}, \tag{21}$$

where $\sigma_{\Delta,\rm seed}$ includes the uncertainty of the acquired frequency and the implemented seed reference. This condition reserves enough detuning margin for the first short-pulse measurement after the blind step. The determination of $\Delta_{\rm blind}^{\max}$ and the choice of $\delta\Phi_{\rm blind}$ are described in Appendix A.

No measured sensitivity is available before this step, so the first short-pulse probe at Φ_1 uses $\omega_{d,1} = \widehat{\omega}_{01,\rm seed}$. Its output supplies the second frequency estimate $\widehat{\omega}_{01,1}$ and hence the initial secant sensitivity

$$\widehat{S}_{\Phi,1} = \frac{\widehat{\omega}_{01,1} - \widehat{\omega}_{01,\rm seed}}{\Phi_1 - \Phi_{\rm seed}}. \tag{22}$$

This initial slope combines the scanned-Ramsey seed with the short-pulse estimate at Φ_1. The uncertainty assigned to $\widehat{S}_{\Phi,1}$ includes the uncertainties of both frequency estimates.

For subsequent iterations, the local sensitivity is estimated from the two most recent TRACK measurements,

$$\widehat{S}_{\Phi,n} = \frac{\widehat{\omega}_{01,n} - \widehat{\omega}_{01,n-1}}{\Phi_n - \Phi_{n-1}}, \qquad n \ge 2. \tag{23}$$

This measurement-derived quantity approximates the local flux-to-frequency sensitivity without supplying the controller with the quantitative transmon dispersion or its derivative. A sensitivity estimate is accepted only when its sign is consistent with χ_Φ and its magnitude and uncertainty satisfy the prescribed reliability thresholds. Failure of this check produces the local-failure event defined in Sec. II.C.

At each accepted TRACK iteration, the controller forms

$$\widehat{r}_n = \widehat{\omega}_{01,n} - \omega_{\rm tar} \tag{24}$$

and tests Eq. (**??**). If that condition is not satisfied, the controller proposes the damped secant–Newton update

$$\Phi_{n+1}^{\mathrm{prop}} = \Phi_n - \eta \frac{\widehat{r}_n}{\widehat{S}_{\Phi,n}}, \tag{25}$$

where $\eta \in (0, 1]$ is the damping factor. Before application, the commanded bias change is capped at $\delta\Phi_{\max}$ in magnitude, and the resulting bias is restricted to $[\Phi_{\min}, \Phi_{\max}]$. The uncertainty propagation, sensitivity thresholds, limiting convention, and controller parameters are specified in Appendix A.

The flux-bias update changes the physical working point and is intended to reduce the residual relative to the fixed target. It does not by itself ensure that the short-pulse probe remains within its validated detuning interval. The drive-reference and range controller that enforces this second requirement is constructed next.

D. Drive-reference tracking and range maintenance

After the flux bias is updated, the detuning of the next measurement is

$$\Delta_{n+1} = \omega_{01}(\Phi_{n+1}) - \omega_{d,n+1}. \tag{26}$$

If the microwave reference is held fixed, $\omega_{d,n+1} = \omega_{d,n}$, this becomes

$$\Delta_{n+1}^{\mathrm{fixed}} = \omega_{01}(\Phi_{n+1}) - \omega_{d,n}. \tag{27}$$

A short-pulse probe referenced to a fixed drive is therefore forced to span the accumulated frequency displacement as the physical bias moves. The calibration trajectory can consequently leave the validated interval even when each flux-bias update moves the qubit toward the target.

During Track, we instead predict the transition frequency at the updated bias using the measured secant sensitivity and set

$$\omega_{d,n+1} = \widehat{\omega}_{01,n} + \widehat{S}_{\Phi,n}\left(\Phi_{n+1} - \Phi_n\right). \tag{28}$$

The right-hand side is the predicted transition frequency at Φ_{n+1} based on measurements available through iteration n. This update uses only estimates already produced by the loop and does not evaluate Eq. (1). Substitution into Eq. (26) shows that the next measurement detuning is the one-step frequency-prediction error rather than the accumulated frequency displacement. Drive tracking therefore reduces the risk of leaving the validated interval when the local secant prediction remains reliable, but it does not by itself guarantee range validity.

Let $\sigma_{\mathrm{pred},n+1}$ denote the uncertainty of the frequency prediction on the right-hand side of Eq. (34). It includes the uncertainties of the current frequency estimate, the secant sensitivity, and the applied bias step. Before the next short-pulse probe is executed, the prediction must satisfy

$$z_{1-\beta}\sigma_{\mathrm{pred},n+1} \le \Delta_{\mathrm{val}}. \tag{29}$$

For the initial blind step, the corresponding admission test is Eq. (21). Failure of either premeasurement test produces a local-failure event and invokes Reacquire before the local inverse is used.

After an admitted short-pulse measurement, the resulting estimate must also satisfy

$$|\widehat{\Delta}_{n+1}| + z_{1-\beta}\sigma_{\Delta,n+1} \le \Delta_{\mathrm{val}}. \tag{30}$$

This postmeasurement condition checks the consistency of the measured detuning with the response-validation interval. Failure of this condition, numerical convergence, inverse-branch continuity, or the sensitivity check rejects the local estimate and returns the workflow to Reacquire.

When the range and estimator checks remain satisfied, the resulting frequency estimate is passed to the next flux-bias update. Satisfaction of Eq. (**??**) instead produces a target-entry event and transfers the workflow to Lock, where calibration success is determined only by the independent verification condition Eq. (9).

The flux-bias and drive-reference updates have different physical roles. Equation (25) changes the qubit working point, whereas Eq. (34) changes only the reference of the next measurement. The drive update neither redefines the target nor removes the calibration residual, because every frequency estimate remains compared with the fixed ω_{tar}. Together, the two updates implement the local Track state defined in Sec. II.C.

IV. CLOSED-LOOP PROTOCOL WITH DRIVE TRACKING

The control objective is to find a flux offset V such that

$$r(V) = \omega_q(V) - \omega_{\mathrm{tar}} = 0. \tag{31}$$

At iteration n, the transient discriminator returns $\widehat{\omega}_{q,n} = \omega_{d,n} + \widehat{\Delta}_n$. The estimated feedback residual is $\widehat{r}_n = \widehat{\omega}_{q,n} - \omega_{\mathrm{tar}}$. A damped gradient update changes the flux bias according to

$$V_{n+1} = \mathrm{clip}\left(V_n - \eta\frac{\widehat{r}_n}{\widehat{S}_n}, V_{\min}, V_{\max}\right), \tag{32}$$

where η is a damping factor and $\widehat{S}_n$ is a local slope. After two measurements, a model-free secant estimate is

$$\widehat{S}_n = \frac{\widehat{\omega}_{q,n} - \widehat{\omega}_{q,n-1}}{V_n - V_{n-1}}. \tag{33}$$

The measurement reference is updated separately from the controlled bias. We predict the qubit frequency at the next point using

$$\omega_{d,n+1} = \widehat{\omega}_{q,n} + \widehat{S}_n(V_{n+1} - V_n). \tag{34}$$

Equation (34) serves only to keep $|\omega_q(V_{n+1}) - \omega_{d,n+1}|$ within the local discriminator window. It does not change the optimization target in Eq. (31). An initial frequency estimate seeds the first drive, and a bounded blind step provides the second point needed for Eq. (33). Subsequent updates use the measured secant slope, so the procedure does not require an analytic frequency–flux model.

The complete protocol therefore has two coupled but distinct loops: the flux update reduces the target error, whereas the drive update reduces the measurement detuning. A range guard can trigger a wide-capture Ramsey measurement if the local discriminator leaves its validated interval. The numerical experiment below isolates the local tracking phase and starts from a frequency seed sufficient to keep the transient probe on its connected branch.

V. NUMERICAL RESULTS

We simulate a two-level flux-tunable transmon with $E_C/h = 0.2$ GHz, $E_J/h = 10$ GHz, $T_1 = 100$ μs, and $T_2 = 50$ μs. The reference flux is $\Phi_{\rm opt} = 0.9553\Phi_0$, the time step is 0.5 ns, and each $\pi/2$ pulse occupies a 10-ns control window. The master equation is solved with QuTiP. Reported frequency errors are independently checked against the analytic transmon dispersion, rather than against the estimate used by the feedback controller.

A. Validity of the local estimator

Figure 2 compares the linear inverse with two cubic calibrations while the drive is fixed at the reference frequency. The full kernel and odd-polynomial routes agree in the central region, providing an independent check of the cubic coefficient. The inferred turnover is $|\Delta| \simeq 17.6$ MHz. Within the sampled points on this branch, the mean absolute error is 1.27 MHz for the linear estimator, 0.94 MHz for the fitted cubic estimator, and 0.97 MHz for the full-kernel estimator. Near the fold, the cubic correction can overshoot and the local inverse ceases to improve the estimate. The result defines an operating window; it does not justify using the transient probe for wide-range capture.

B. Accuracy–cost tradeoff

To separate estimator cost from fold failure, we next restrict the flux offsets to $\pm 0.01\Phi_0$, corresponding to $|\Delta| \lesssim 12$ MHz. Figure 3 compares six strategies. Double-sweep Ramsey reaches a 6.3-kHz mean absolute error using 800 master-equation solves per frequency measurement. Single-sweep Ramsey configurations use 100–400 solves and give 23–30-kHz errors. The cubic transient estimator gives a 13.5-kHz error at an amortized cost of 18 solves per point. Its marginal cost is two solves after a one-time 80-solve coefficient calibration. In this fixed-drive benchmark, the cubic transient point Pareto-dominates all three single-sweep Ramsey settings; double-sweep Ramsey remains more accurate at substantially higher cost.

C. Closed-loop calibration

We set the target frequency 20 MHz below the reference point and use the same initial flux offset for all strategies. Figure 4 contrasts three feedback loops. A conventional double-sweep Ramsey loop converges in five iterations. A coarse transient stage followed by Ramsey refinement takes seven and five iterations, respectively; the transient stage begins outside its fold window and wanders before the handoff. The drive-tracked transient loop instead converges monotonically in six iterations without a Ramsey refinement stage.

Raw iteration count is not a fair cost metric because one Ramsey measurement contains a delay scan. We therefore instrument calls to both the master- and Schrodinger-equation solvers and count each call with unit weight. This is conservative for the transient method because the state-vector solver used for kernel evaluation is cheaper than the master-equation solver. As shown in Fig. 5, the drive-tracked loop uses 984 counted calls and reaches a verified residual of 0.00519 MHz. Ramsey alone uses 4000 calls and reaches 0.01797 MHz, giving a 4.1-fold reduction in counted cost. The coarse-to-fine workflow uses 5214 calls despite its 0.00214-MHz final residual; at the common 0.016-MHz tolerance its speedup relative to Ramsey is only 0.8, so it is more expensive rather than faster.

The mechanism is visible directly in the measured detuning. After bootstrap, drive tracking predicts the next transition frequency closely enough that the probe detuning falls from sub-megahertz values to a few kilohertz while the absolute target residual spans tens of megahertz. The inexpensive probe thus remains local even though the calibration travels over a nonlocal frequency interval. By contrast, keeping the drive fixed forces the probe itself to span the full target displacement and crosses the turnover identified in Fig. 2.

VI. DISCUSSION

The numerical results answer the motivating question conditionally: short pulse sequences can contribute directly to frequency calibration, but they do not automatically become general-purpose frequency estimators. The present two-branch sequence is a local probe, not a wide-range replacement for Ramsey spectroscopy. Its cubic response reduces local nonlinear bias but cannot remove a response fold. In closed loop, the useful range

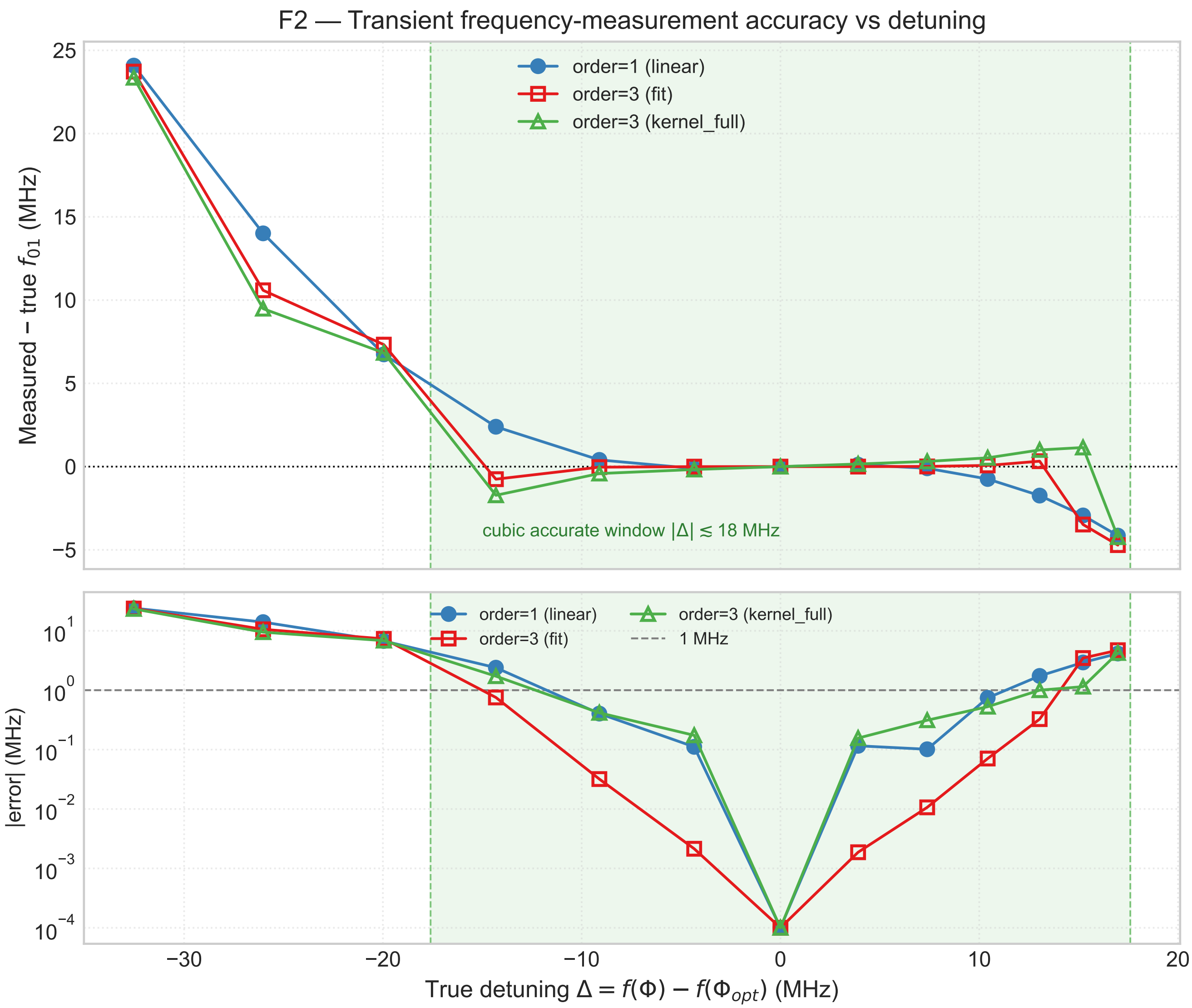

FIG. 2. Accuracy of the transient frequency discriminator as a function of true detuning. The upper panel shows signed measurement error and the lower panel its magnitude. The shaded interval marks the cubic branch bounded by the response turnover. The fixed-drive polynomial fit and full three-time kernel give consistent cubic corrections in the central region.

is preserved by changing the measurement reference so that the short-pulse probe stays local; merely assigning it to a coarse stage before an expensive fine estimator does not guarantee a lower total cost.

This framing clarifies what it means for a pulse sequence to participate in calibration. The sequence must encode the parameter in an observable response, that response must admit a calibrated inverse over a stated operating window, and the surrounding protocol must keep subsequent measurements inside that window. In the present example, the qubit experiences constant detuning throughout two finite pulses, so the third-order coefficient contains unequal-time contributions. A time-diagonal kernel is not the relevant object. An odd-polynomial scan can recover the same coefficient when centered at the symmetry point, but that assumption is fragile under a moving drive. Evaluating the full response kernel at the actual reference frequency makes the short-pulse estimator and controller definitions consistent.

The cost comparison is intentionally based on instrumented solver calls rather than a nominal workflow price. The nominal price counts the two population measurements but can omit the solver work required to recalibrate a response coefficient when the drive changes. Counting every solver invocation avoids a large artificial speedup. Wall-clock time supports the same ordering in the present implementation, but it depends on hardware, caching, and solver choice and is not a universal physical metric.

Several steps are needed before interpreting the proto-

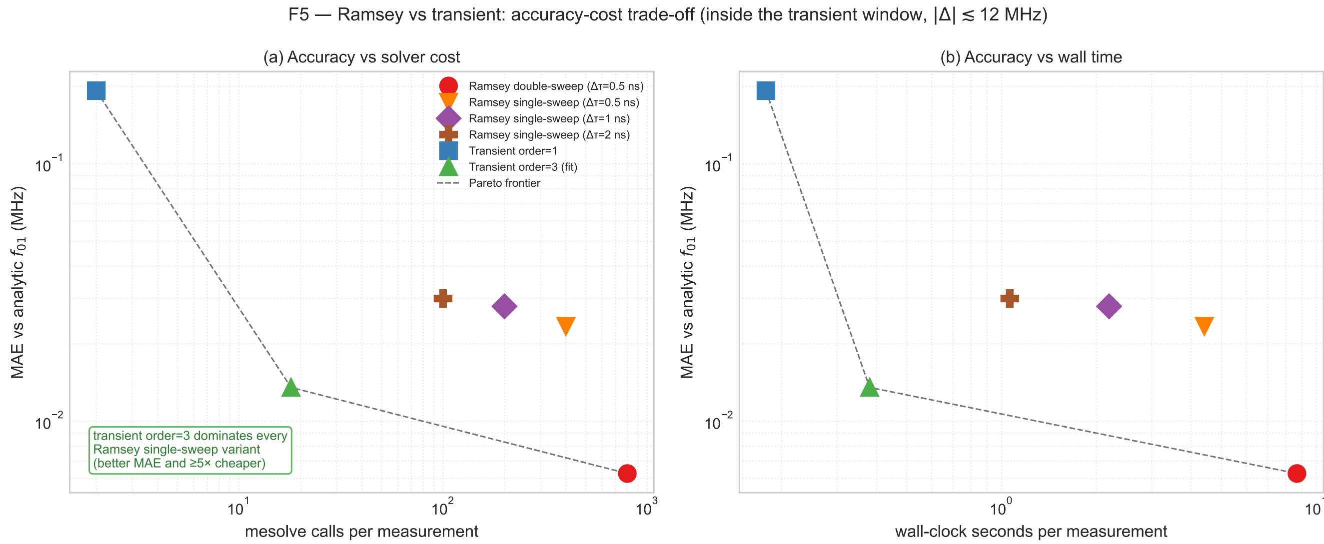

FIG. 3. Accuracy–cost comparison inside the local transient window. (a) Mean absolute frequency error versus measured master-equation calls. (b) Error versus wall-clock time on the benchmark machine. Dashed lines mark the nondominated frontier. Transient cubic cost includes amortization of its one-time coefficient calibration over five flux points.

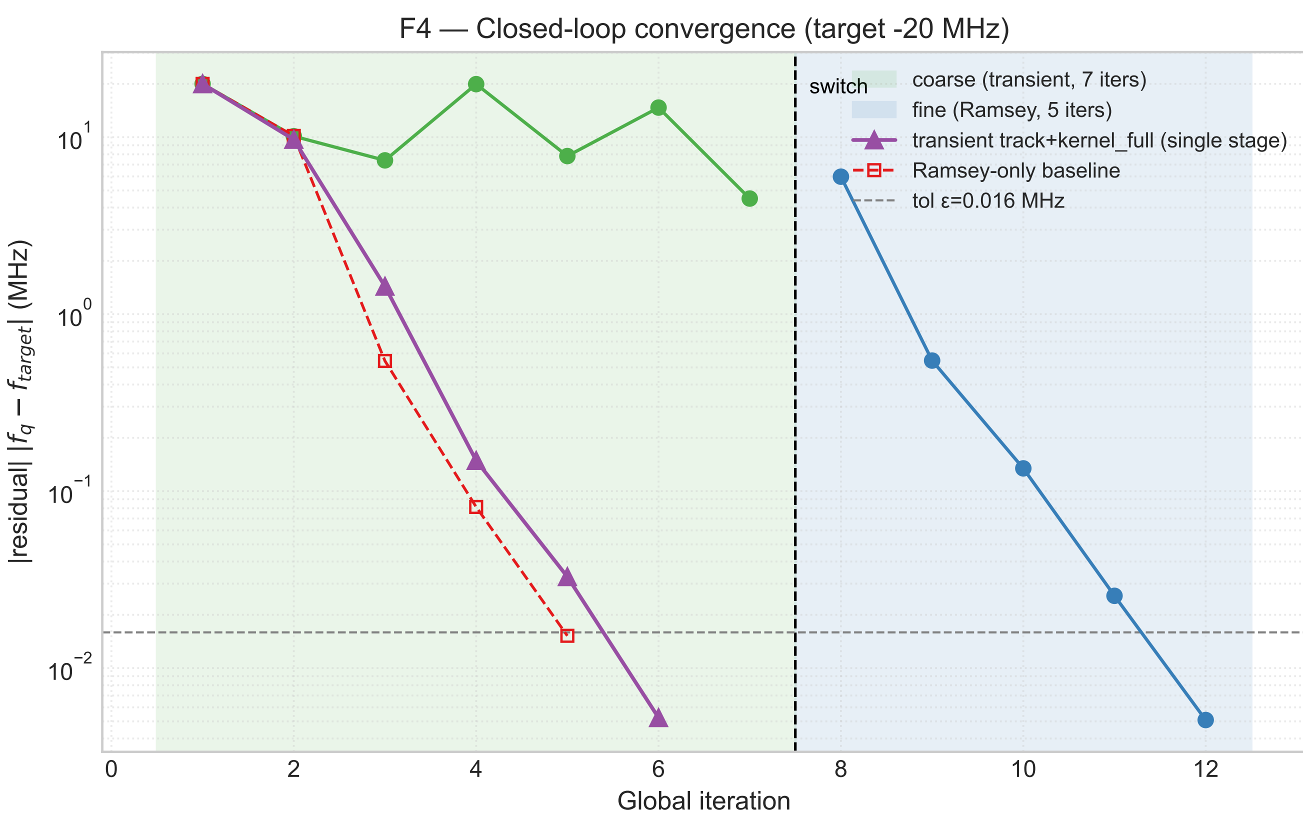

FIG. 4. Closed-loop residual for a target shift of −20 MHz. The coarse-to-fine workflow (green and blue regions) is compared with the drive-tracked full-kernel protocol (purple triangles) and a Ramsey-only baseline (red squares). Residuals are independently evaluated from the analytic frequency–flux relation.

col as an experimental calibration advantage. The current simulations use ideal control envelopes, projective

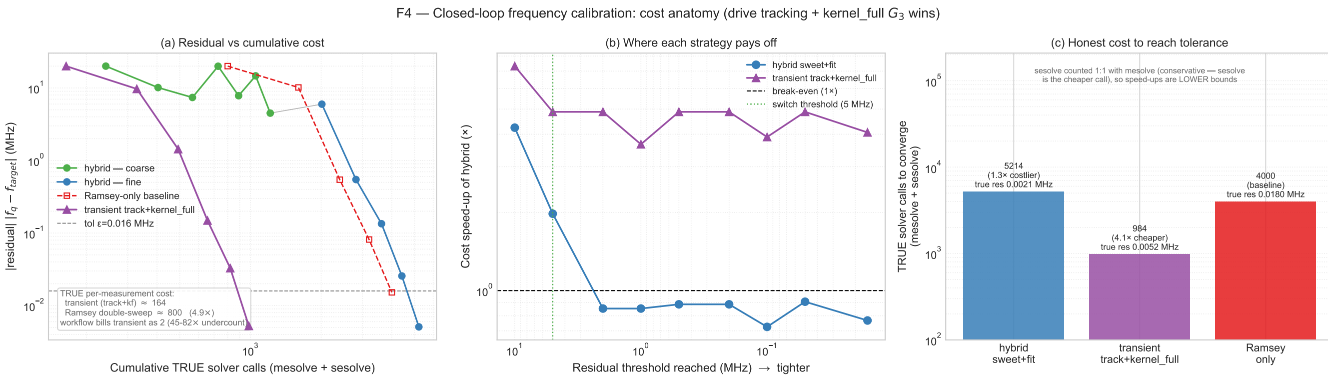

FIG. 5. Cost anatomy of closed-loop frequency calibration. (a) Verified residual versus cumulative instrumented solver calls. (b) Cost ratio relative to the Ramsey baseline at a common residual threshold. (c) Total counted calls and independently verified final residuals. The full-kernel transient route uses drive tracking; the hybrid route uses a fixed reference in its transient stage.

readout, a Markovian two-level model, and deterministic expectation values. Shot noise, assignment error, drift during acquisition, pulse distortion, leakage, and imperfect knowledge of the kernel will perturb both the detuning estimate and the secant slope. An experiment should report the number of circuit repetitions and elapsed acquisition time, and it should recompute the cost-to-reach curve at a common residual threshold. A wide-range capture or range-guard path is also required when no reliable initial frequency seed is available.

These limitations suggest a practical hierarchy rather than a replacement claim. Spectroscopy or a Ramsey scan supplies acquisition and recovery. Once the transition frequency is known, short-pulse sequences can participate in frequent local maintenance, here through a drive-tracked two-branch discriminator. Occasional wide-range measurements can reinitialize the local loop after a range violation. This division uses pulse-time information where it is unambiguous while retaining established estimators for global capture.

VII. CONCLUSION

We have investigated whether short-pulse sequences can participate directly in superconducting-qubit frequency calibration. In the example studied here, a two-branch sequence provides a local detuning discriminator without a free-evolution scan. A full third-order response kernel improves its local inverse, while a drive reference that tracks the predicted qubit frequency prevents the feedback trajectory from exhausting that local validity. In the simulated 20-MHz tuning task, the protocol reduces the instrumented solver cost by a factor of 4.1 relative to double-sweep Ramsey and reaches a smaller independently checked residual. A coarse short-pulse stage followed by Ramsey refinement does not provide the same advantage.

The result demonstrates a qualified route for bringing short-pulse dynamics into calibration: identify a parameter-sensitive pulse response, calibrate its local inverse, and preserve that inverse's validity as the device moves. It does not establish that short sequences replace spectroscopy or Ramsey for global acquisition. Experimental validation with finite shots, readout error, drift, and control imperfections will determine how broadly this role extends and whether it reduces acquisition time on hardware.

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Appendix A: Numerical parameters and cost accounting

The parameters used for the frozen numerical results are summarized in Table I. Frequencies in the simulation are angular frequencies; reported errors are divided by 2π and expressed in megahertz.

TABLE I. Parameters of the numerical calibration benchmark.

Parameter	Value
E_C/h	0.2 GHz
E_J/h	10 GHz
Reference flux	$0.9553166\Phi_0$
T_1	100 μs
T_2	50 μs
Hilbert-space dimension	2
Sampling interval	0.5 ns
$\pi/2$ pulse window	10 ns
Closed-loop target shift	-20 MHz
Residual tolerance	0.016 MHz
Damping factor	0.8
Maximum flux step	$0.02\Phi_0$

For the accuracy–cost benchmark, Ramsey free-evolution times span $0 \leq \tau < 200$ ns with spacings of 0.5, 1, or 2 ns. Double-sweep Ramsey uses two artificial-detuning scans; single-sweep Ramsey uses one. The transient measurement requires two final-state simulations. The fitted cubic route pays an additional one-time calibration cost, which is amortized across the five flux offsets in Fig. 3.

The closed-loop benchmark instruments all calls made through the frequency measurement and kernel-estimation paths. Calls to the master-equation solver and the state-vector solver are assigned equal unit cost. The latter is typically cheaper, so the reported transient advantage in solver-call units is conservative. Final residuals are recomputed from the analytic transmon dispersion at the returned flux point and are not taken from the internal measurement history.